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ASSIGNMENT: 8

1. **Write a Program to perform the following operations on the array:**

Algorithm:

1. Start
2. Input size and elements of array
3. Pass array and size to a function by reference
4. In the function, multiply each element by 2
5. Return to main
6. Print modified array
7. Stop

Code:

```
#include <stdio.h>


void multiplyByTwo(int *arr, int n) {
    for(int i = 0; i < n; i++) {
        arr[i] *= 2;
    }
}


int main() {
    int arr[10], n;
    printf("Enter number of elements: ");
    scanf("%d", &n);


    printf("Enter elements: ");
    for(int i = 0; i < n; i++)
        scanf("%d", &arr[i]);
```

```
    multiplyByTwo(arr, n);

    printf("Modified array: ");
    for(int i = 0; i < n; i++)
        printf("%d ", arr[i]);
    return 0;
}
```

2. **Write a C program that searches for an element in an array using a user defined function and pointers.**
   Input: Enter elements: 10 20 30 40 50 60
   Output: Enter element to search: 30
   Element found at index: 2
   Algorithm:
   1. Start
   2. Input array and element to search
   3. Pass array, size, and target to a function using pointers
   4. In function, loop through and compare
   5. Return index if found
   6. Else return -1
   7. Print result
   8. Stop

   Code:

```
#include <stdio.h>

int searchElement(int *arr, int n, int key) {
    for(int i = 0; i < n; i++) {
        if(*(arr + i) == key)
            return i;
    }
    return -1;
```

```
}

int main() {
    int arr[100], n, key;
    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements: ");
    for(int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    printf("Enter element to search: ");
    scanf("%d", &key);

    int result = searchElement(arr, n, key);
    if(result != -1)
        printf("Element found at index: %d\n", result);
    else
        printf("Element not found\n");

    return 0;
}
```

3. **Write a C program that calculates the factorial of a number using a function.**

   Algorithm:

   1. Start
   2. Input number
   3. Call factorial function

4. Calculate factorial using loop
5. Return result
6. Print factorial
7. Stop

Code:

```
#include <stdio.h>


int factorial(int n) {
   int fact = 1;
   for(int i = 1; i <= n; i++)
      fact *= i;
   return fact;
}


int main() {
   int n;
   printf("Enter a number: ");
   scanf("%d", &n);


   printf("Factorial: %d\n", factorial(n));
   return 0;
}
```

4. **Write a C program to remove spaces in a string using pointers.**

Algorithm:

1. Start
2. Input string
3. Use pointer to copy non-space characters
4. Print modified string
5. Stop

Code:

```
#include <stdio.h>

void removeSpaces(char *str) {
    char *p = str, *q = str;
    while(*p) {
        if(*p != ' ')
            *q++ = *p;
        p++;
    }
    *q = '\0';
}

int main() {
    char str[100];
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    removeSpaces(str);
    printf("String without spaces: %s\n", str);
    return 0;
}
```

5. **Write a C program to replace a character in a string by another character using pointers**
   Test case:
   Welcome to VIT
   O/p: Welcxme tx VIT

Algorithm:

1. Start
2. Input string, character to replace, and replacement
3. Traverse string using pointer
4. Replace where character matches
5. Print modified string
6. Stop

Code:

```
#include <stdio.h>


void replaceChar(char *str, char oldChar, char newChar) {
    while(*str) {
        if(*str == oldChar)
            *str = newChar;
        str++;
    }
}


int main() {
    char str[100];
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);


    replaceChar(str, 'o', 'x');
    printf("Modified string: %s\n", str);
    return 0;
}
```

6. **Write a program to create a user defined function to join two strings without using Library function and print concatenated strings on the screen.**
   String1: Ram
   String2: Sharma
   Stirng1: Ram Sharma
   Algorithm:
   1. Start
   2. Input both strings
   3. Use a loop to reach end of first string
   4. Append second string character by character
   5. Add null terminator
   6. Print result
   7. Stop

   Code:

```
#include <stdio.h>

void concat(char *s1, char *s2) {
    while(*s1) s1++;
    *s1++ = ' '; // add space
    while(*s2)
        *s1++ = *s2++;
    *s1 = '\0';
}

int main() {
    char s1[100] = "Ram";
    char s2[] = "Sharma";

    concat(s1, s2);
    printf("String1: %s\n", s1);
    return 0;
```

```
}
```

7. **Write a program to create a user defined function for finding length of String and print on the screen without using library function.**

Algorithm:

1. Start
2. Input string
3. Count till null terminator
4. Print count
5. Stop

Code:

```
#include <stdio.h>


int stringLength(char *str) {
   int len = 0;
   while(*str++) len++;
   return len;
}


int main() {
   char str[100];
   printf("Enter a string: ");
   fgets(str, sizeof(str), stdin);


   printf("Length: %d\n", stringLength(str));
   return 0;
}
```

8. **Write a program to compare the contents of two strings whether strings are equal or not and print appropriate messages on the screen without using the library function.**

Algorithm:

1. Start
2. Input two strings
3. Compare char by char
4. If mismatch, not equal
5. If both end, equal
6. Stop

Code:

```
#include <stdio.h>


int compareStrings(char *s1, char *s2) {
   while(*s1 && *s2) {
      if(*s1 != *s2)
         return 0;
      s1++; s2++;
   }
   return (*s1 == '\0' && *s2 == '\0');
}


int main() {
   char s1[100], s2[100];
   printf("Enter first string: ");
   fgets(s1, sizeof(s1), stdin);
   printf("Enter second string: ");
   fgets(s2, sizeof(s2), stdin);

   if(compareStrings(s1, s2))
      printf("Strings are equal.\n");
```

```
    else
        printf("Strings are not equal.\n");
    return 0;
}
```

9. **Given a string s, perform following operations on a string using function (Without pointer):**
    a. **Find a length of a string.**
    b. **Print string in reverse order.**
    c. **Copy string s into s1 and print it.**
    d. **Accept another string, say s2. Concatenate s and s2 and print concatenated string.**

Code:

```
#include <stdio.h>

int strLength(char str[]) {
    int i = 0;
    while(str[i] != '\0' && str[i] != '\n') i++;
    return i;
}

void strReverse(char str[]) {
    int len = strLength(str);
    for(int i = len - 1; i >= 0; i--)
        printf("%c", str[i]);
    printf("\n");
}

void strCopy(char src[], char dest[]) {
    int i = 0;
    while(src[i]) {
        dest[i] = src[i];
        i++;
    }
    dest[i] = '\0';
}
```

```
void strConcat(char s1[], char s2[]) {
    int i = strLength(s1);
    s1[i++] = ' ';
    int j = 0;
    while(s2[j]) {
        s1[i++] = s2[j++];
    }
    s1[i] = '\0';
}

int main() {
    char s[100], s1[100], s2[100];
    printf("Enter string: ");
    fgets(s, sizeof(s), stdin);

    printf("Length = %d\n", strLength(s));
    printf("Reverse = ");
    strReverse(s);

    strCopy(s, s1);
    printf("Copied string: %s\n", s1);

    printf("Enter second string: ");
    fgets(s2, sizeof(s2), stdin);
    strConcat(s, s2);
    printf("Concatenated: %s\n", s);
    return 0;
}
```

10. Write a program to check whether a string is a palindrome.

Algorithm:

1. Start
2. Input string
3. Reverse and compare with original
4. If match, it's a palindrome
5. Print result
6. Stop

Code:

```
#include <stdio.h>
```

```
int isPalindrome(char *str) {
    int len = 0;
    while(str[len] && str[len] != '\n') len++;

    for(int i = 0; i < len / 2; i++) {
        if(str[i] != str[len - i - 1])
            return 0;
    }
    return 1;
}

int main() {
    char str[100];
    printf("Enter string: ");
    fgets(str, sizeof(str), stdin);

    if(isPalindrome(str))
        printf("Palindrome\n");
    else
        printf("Not Palindrome\n");

    return 0;
}
```